MIDTERM COMPREHENSIVE STUDY GUIDE

SEN4100 – Software Project Management | Fall Semester (Weeks 1–6)

1. Foundations of Software Project Management (Week 1)

- **Project vs. Operations:** A project is a temporary endeavor to create a unique product or result. Operations are continuous, repetitive processes sustaining business continuity.
- **The Intangibility of Software:** Software projects are uniquely difficult because progress is invisible, changes appear superficially easy (Illusion of Malleability), and software does not wear out mechanically—it degrades via technical debt.
- **The CHAOS Report:** Software projects fail primarily due to managerial and communication issues (volatile requirements, politically imposed deadlines), not technical incompetence.
- **Catastrophic Case Studies:**
  - *Ariane 5 Flight 501:* Loss due to an unverified 64-bit to 16-bit integer overflow. Taught that components cannot be reused without system-level validation.
  - *Healthcare.gov:* Systemic collapse due to extreme vendor fragmentation, lack of end-to-end testing, and fixed deadlines with expanding scope.
- **Constraint Paradigms:**
  - *Classic Iron Triangle:* Scope, Time, Cost (Vertices); Quality is the dependent variable.
  - *Expanded PMBOK Vector:* Adds Risk Exposure, Resource Availability, and Business Value.
- **Lifecycle Duality:** Project Life Cycle (PLC) represents the governance container (Initiation → Closure). Software Development Life Cycle (SDLC) represents the technical flow (Requirements → Deployment).

2. Process Models & Lifecycles (Week 2)

- **Generic Process Framework:** 5 universal activities (Communication, Planning, Modeling, Construction, Deployment) supported by Umbrella Activities (QA, SCM, Risk Management).
- **Predictive (Plan-Driven) Models:**
  - *Classic Waterfall:* Linear progression. Requires frozen requirements. High risk of "Big Bang" integration failures late in the project.
  - *V-Model:* Pairs every construction phase with a testing phase. Relies on Bidirectional Traceability. Ideal for safety-critical systems (DO-178C, FDA).

- **Exploratory & Risk-Driven Models:**
  - *Prototyping:* Used for ambiguous requirements. *Throwaway* (discarded) vs. *Evolutionary* (built into production).
  - *Spiral Model (Boehm):* 4 quadrants driven by Risk Evaluation. Uses radial distance (cost) and angular dimension (progress). Forces a go/no-go risk review per loop.
- **The Agile Paradigm:**
  - *Inverting the Iron Triangle:* Traditional models fix Scope and estimate Time/Cost. Agile fixes Time (Sprints) and Cost (Team), making **Scope** the variable parameter.
  - *Manifesto Core:* Values individuals, working software, customer collaboration, and responding to change.

3. Project Scope, Feasibility & Initiation (Week 3)

- **Scope Duality:**
  - *Product Scope:* Functional/non-functional features of the software (measured against SRS).
  - *Project Scope:* The work, governance, and infrastructure required to deliver it (measured against WBS).
- **TELOS Feasibility:** Evaluates Technical, Economic, Legal, Operational, and Schedule viability.
- **Economic Feasibility (NPV):** The definitive metric accounting for the time value of money. If NPV > 0, the project is financially viable.
  $NPV = \sum [CF_t / (1 + r)^t] - I_0$
- **Stakeholder Governance (Mendelow's Matrix):** Maps stakeholders by Power and Interest. High Power/High Interest = *Manage Closely*. High Power/Low Interest = *Keep Satisfied*.
- **Project Charter:** The foundational document that formally authorizes the project and vests the Project Manager with resource-allocation authority.
- **Scope Creep & Change Control:** Uncontrolled expansion of scope. Must be governed by a Change Control Board (CCB) evaluating technical and financial impact before updating baselines.

4. WBS & Cost/Effort Estimation (Week 4)

- **Work Breakdown Structure (WBS):** A hierarchical, deliverable-oriented decomposition of scope.
- **The 100% Rule:** The WBS must capture 100% of the project scope (including PM and QA). If it's not in the WBS, it has no budget and will not be built.
- **The Work Package:** The lowest assignable node in the WBS.
  - *8/80 Rule:* A task should take between 8 and 80 hours.
  - *WBS Dictionary:* Details exact specs, owners, and acceptance criteria to prevent "Gold Plating".

- **Software Sizing:**
  - *Lines of Code (LOC):* Easy to measure but language-dependent and punishes concise code.
  - *Function Points (FP):* Measures user business value across 5 domains (Inputs, Outputs, Inquiries, Internal Files, External Interfaces). Independent of programming language.
- **Algorithmic Cost Modeling (COCOMO):** Converts size into Effort (Person-Months) and Duration (Months).
  $E = a_b \times (KLOC)^{b_b}$
  $D = c_b \times (E)^{d_b}$
- **Agile Estimation:** Uses *Story Points* (relative sizing based on Volume, Complexity, Uncertainty via Fibonacci sequence) and *Planning Poker* to eliminate Anchoring Bias.

5. Project Scheduling & Risk Management (Week 6)

- **Activity Sequencing (PDM):** Finish-to-Start (FS) is the most common dependency. Others: SS, FF, SF.
- **Critical Path Method (CPM):**
  - *Forward Pass:* Calculates Early Start (ES) and Early Finish (EF).
  - *Backward Pass:* Calculates Late Start (LS) and Late Finish (LF).
  - *Slack (Float):* `LS - ES`. Time a task can be delayed without pushing the deadline.
  - **The Critical Path:** The longest continuous path. Tasks on this path have **exactly zero slack**.
- **PERT (Program Evaluation & Review Technique):** Uses probabilistic estimation: Optimistic (O), Most Likely (M), Pessimistic (P).
  $\text{Expected Duration } (t_e) = (O + 4M + P) / 6$
- **Software Risk Taxonomy:** Technical, Project Management, Organizational, and External. A risk is an *uncertain future event*; if it has already happened, it is an *Issue*.
- **Probability-Impact (P-I) Matrix:** Calculates Risk Exposure (`RE = Probability × Impact`).
- **Risk Response Strategies:** 1) Mitigate, 2) Avoid, 3) Transfer, 4) Accept.

> **Exam Tip:** Be prepared to draw an Activity-on-Node (AON) network, calculate Early/Late dates to find the Critical Path, and mathematically apply the NPV, COCOMO, and PERT formulas.